

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Section 9
Gradient liquid cooling system

Contents

1.	TOOLS AND MATERIAL	3
2.	INSTALLATION	4
3.	FILLING PROCEDURE	9
4.	CLEAN STRAINERS (FILTERS)	11
5.	START UP PROCEDURE	12
6.	APPENDIX: SETPOINT OF SECONDARY TEMPERATURE SWITCHES.....	13

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T	
	Power	Power	Master

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1. TOOLS AND MATERIAL

Delivered with the heat exchanger:

- 1 x 15 meter (blue) hose (gradient coil cooling, supply)
- 1 x 15 meter (yellow or red) hose (gradient coil cooling, return)
- 2 x 5 meter hoses (facility cooling)
- 4 hose adapters
- 6 hose clamps
- 2 quick-connect assemblies
- Deionisation cartridge
- Teflon tape.
- Open end spanner 34 mm
- Open end spanner 1"
- Drain hose 1 meter.
- Funnel

To be organized locally:

25 liter fluid container filled with distilled water and, when possible, provided with a tap-outlet.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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2. INSTALLATION

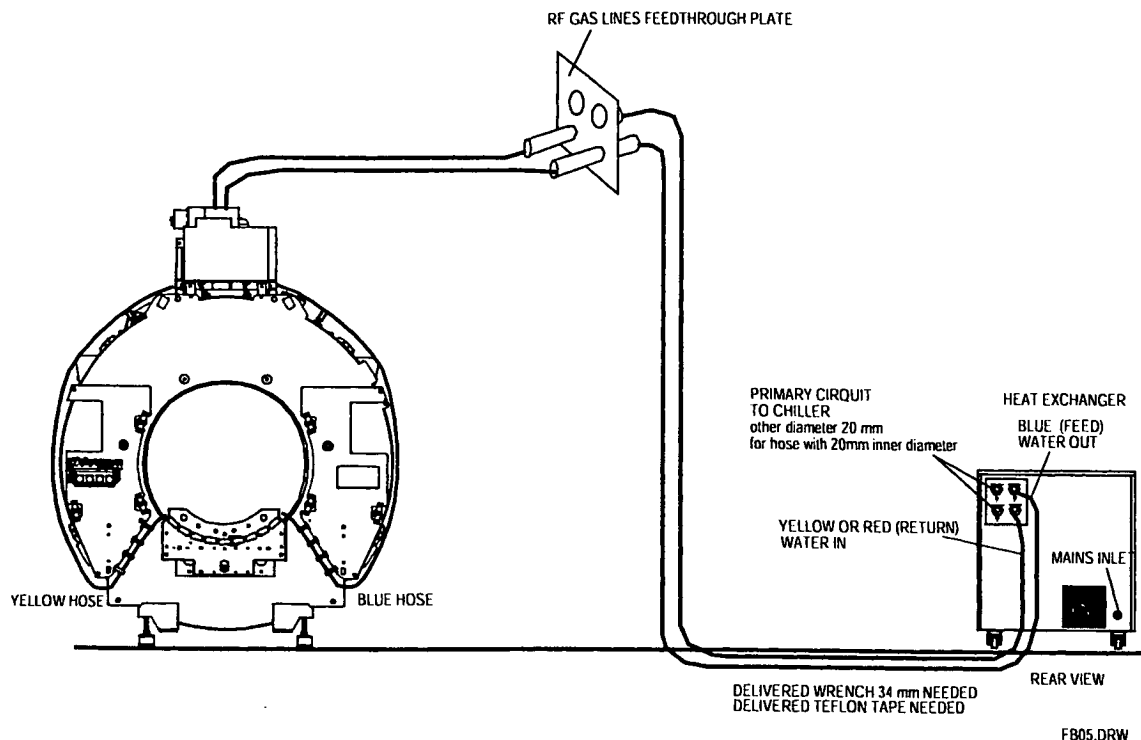
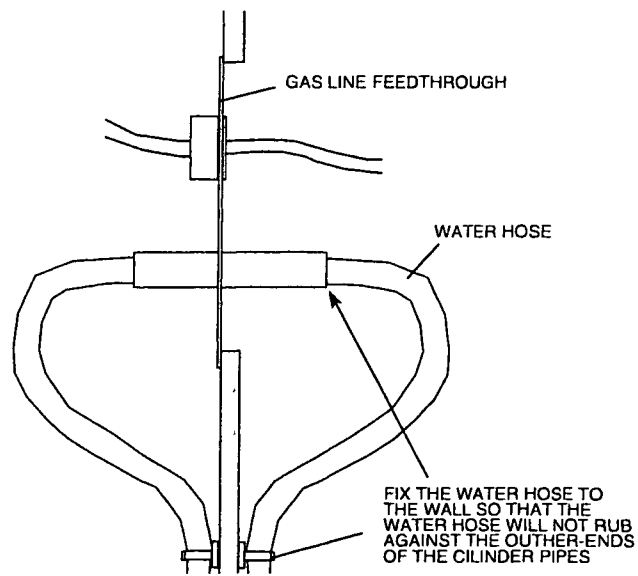


Figure 1 - Installation of the Heat exchanger

1. Connect the water-hose adaptors to 'facility' (= the 'hospital cooling water supply' side) and to 'coil side' of the heat exchanger. An open end spanner of 34 mm is delivered. Put the white teflon tape around the thread of the couplings before mounting.
Note: Keep the length of the hoses, as it is.
2. Lead the two 15 meter water-hoses (coil side) through the cylinders in the RF gas line feedthrough plate.
3. Relieve the hoses, so the sharp-edge of the RF gas line feedthrough filter doesn't damage the hoses.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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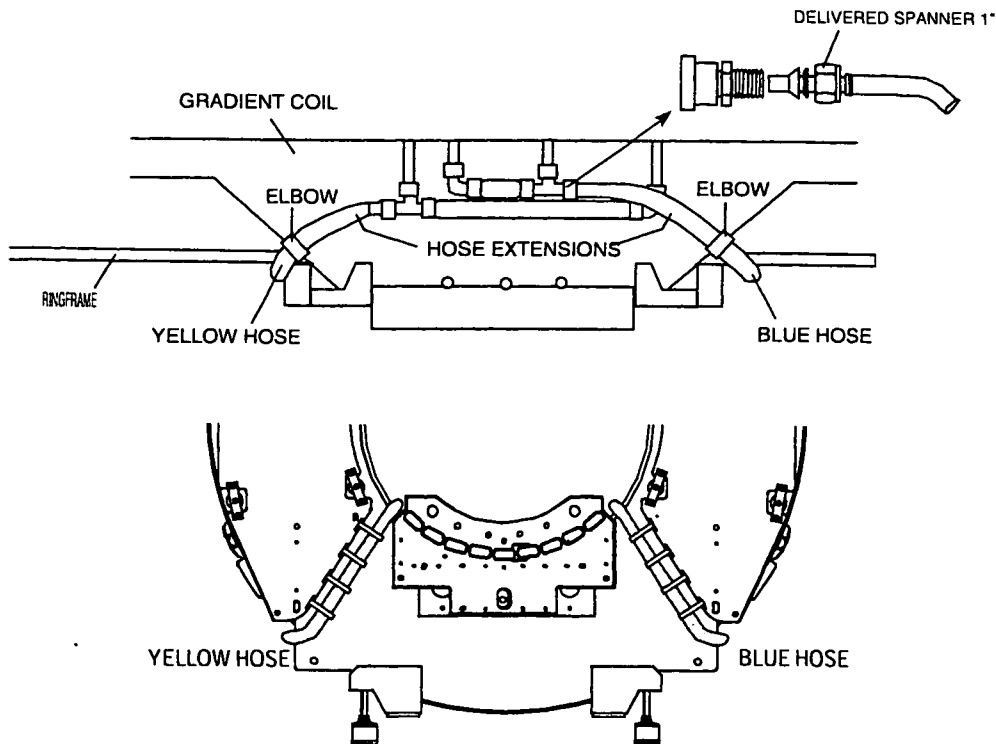


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Figure 2 - Relieve the hoses

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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4. Connect the blue water-hose to the gradient coil-inlet 'FEED' and the yellow or red water-hose to the gradient coil-outlet 'RETURN'. Mount the two Parker fluid fittings as follows:
Ensure that the end of the tube is not damaged. Insert the tube into the coil fitting ensuring that the tube is totally pushed into the coil fitting and mount the nut finger tight. Mark the nut in this position, and tighten the nut for 0.5 to 0.75 turns from the marked position.
The connection is then completed.



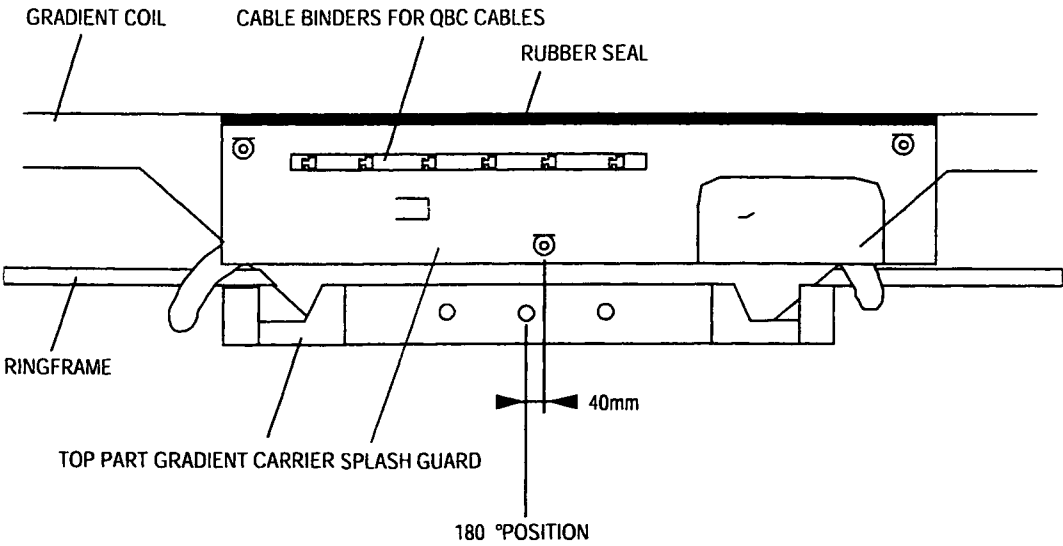
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Figure 3 - Connect hoses to gradient coil

5. Connect the blue water-hose to the heat exchanger coil-outlet 'FEED' and the yellow or red water-hose to the coil-inlet 'RETURN'. Do not forget to use the hose-clamps.
6. Connect both 5 meter water-hoses between the stop valves of the 'hospital cooling water supply' and the corresponding heat exchanger 'facility' adapters. Do not forget to use the hose-clamps.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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7. A 'SPLASH GUARD' must be mounted to protect the Hybrid board from getting wet in case of a leak in one of the hoses or couplings. Mount the splash guard with three screws. These screws are already mounted in the gradient coil.



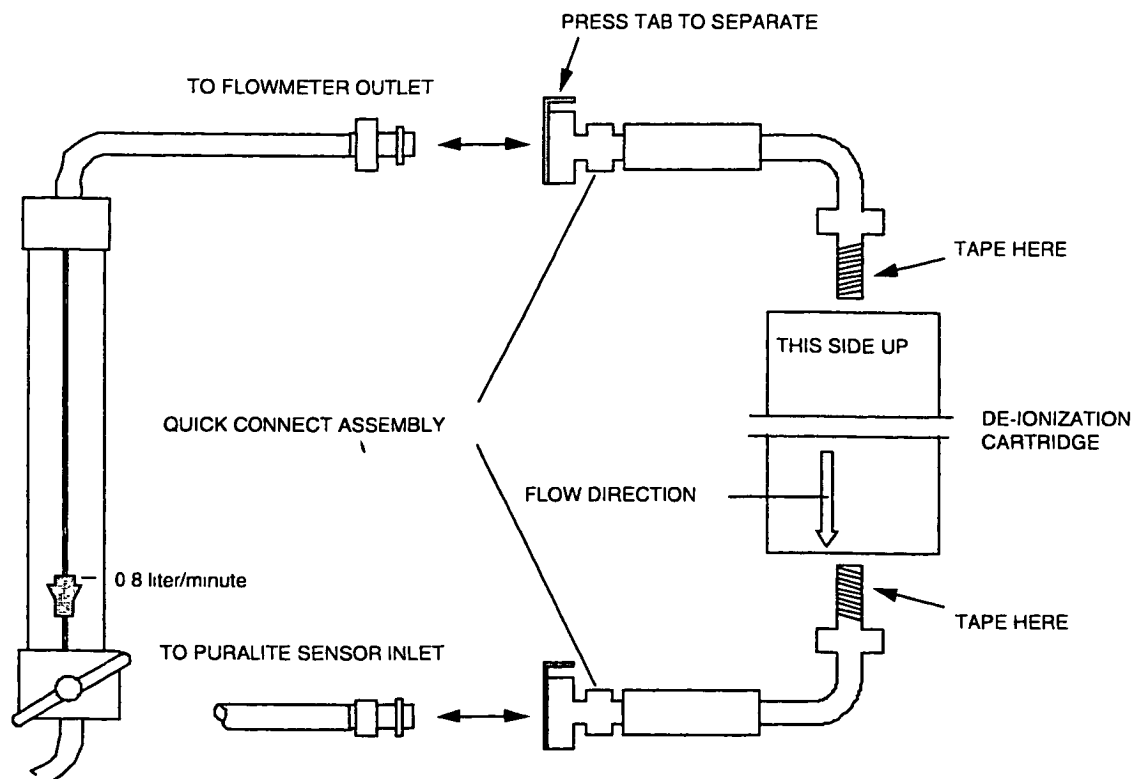
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Figure 4 - Mount the slash guard

After you have mounted the splash guard, fix all cables coming from the QBC and Hybrid board to the splash guard using cable binders.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

8. Install the deionisation cartridge as follows:



HH85 WPG

Figure 5 - Installation of the de-ionisation cartridge.

1. Open the access door on the front of the unit.
2. Separate both quick-connects on the small short-circuit hose by depressing the metal tab on the quick-connect assembly.
3. Remove the plastic bag from the supplied cartridge.
4. Place teflon tape on the threaded ends of the two supplied quick-connect assemblies. Mount the assemblies into the new cartridge.
5. Mount the new cartridge with the 'THIS SIDE UP' sign' upwards behind the door.
6. Connect both quick-connects to the cartridge so that the flow is in the direction as indicated on the de-ionisation filter. Note that the flow direction on the de-ionisation filter can also be from bottom to up (so opposite direction as shown on the figure). In that case connect the output of the flow meter to the bottom of the de-ionisation cartridge.
7. Continue with the filling procedure

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

3. FILLING PROCEDURE

1. Check the mains connections and voltage 220V 50/60 Hz. Also check if the hoses are properly connected.
2. Switch on F7 in the MDU.
3. Turn the TEMPERATURE CONTROL knob on the control panel to 25°C.
4. Remove the reservoir cover from the heat exchanger (plate on top).
5. Locate the reservoir plug (square nut). Remove the plug.
6. Place the fluid container on top of the heat exchanger, with the tap-outlet above the reservoir inlet.
7. Fill the reservoir with distilled water, till the bottom of the fill hole-thread is reached.
Since the reservoir capacity is small, remaining fluid is used to keep the system topped up when pump circulation starts.
8. Switch on the heat exchanger by pressing the POWER switch
The pump starts and the water-level drops.
9. When the reservoir is almost empty, switch off the HE again.
10. Fill again the reservoir till the bottom of the fill hole-thread is reached.
11. Switch on the HE again.
12. Repeat from 9 until the water level stops just below the top while the HE is running.

NOTE

The low fluid level detector watches the water-level. When the level is below safe operating level the low level light lits and the head exchanger will switch off.

CAUTION

After (re)filling the coil circuit, the water purity (red) light will lit for a couple of days. This indicates that the ion-level in the liquid is not yet within specification. The light should change to green within one week. Note that during this first week, it is allowed to operate the system with the red light on. When, after normal operation, the light changes to steady red, the deionisation cartridge should be replaced. If the light is periodically changing red/green, the purity is still adequate but replacement should be anticipated.

11. Mount the reservoir plug by applying teflon tape to provide a good seal.
12. Check the reservoir water level after 3 days. It can take one or two days until all air is out of the system.
13. Install the top cover
14. After initial installation, the strainers become clogged with debris and scale within the first week.
Therefore, the strainers (filters) must be cleaned after one week of operation.
Replace also the de-ionisation cartridge, although the indicator will be still green.
15. Check the flow through the flow meter if you have a unit with flowmeter (spec. 0,8 liter/minute)

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

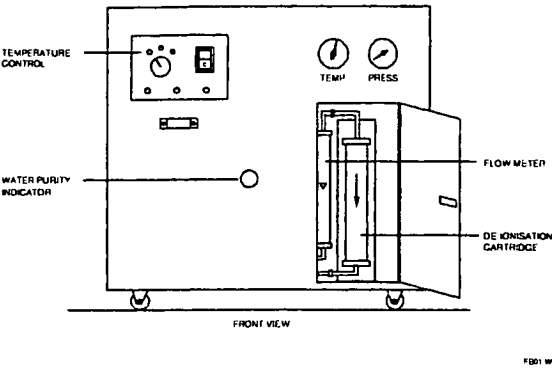


Figure 6 - Location of the de-ionisation cartridge

NOTE

There are also heat exchangers delivered without a flow meter, the flow through the dei-loop is factory adjusted for these units.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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4. CLEAN STRAINERS (FILTERS)

The strainers (filters) must be cleaned after one week of operation after installation. This is to prevent system shutdowns due to low flow errors caused by debris in the strainers.

Material:
Distilled water.
Extra drain-hose
Small reservoir

Procedure:

1. Switch off heat exchanger with F7 in the MDU.
2. Remove the heat exchanger cover. Screws are located in the side covers.
3. Cover both pump motors with a piece of plastic sheet
4. Remove the reservoir plug. A little bit of water may come out till the reservoir pressure is decreased.
5. Remove with the remaining water out of the reservoir using the delivered drain hose.
Place a small reservoir on the floor next to the H E.
6. Fill the drain-hose with water and close the hose on both ends, with your fingers.
7. Put one side of the drain-hose into the heat exchanger reservoir and the other side into the small reservoir.
8. Release the hose-ends and the water flows in the small reservoir.
9. Remove the strainers by unscrewing their housing, two strainers (plastic) are located on the outlet line of the heat exchanger reservoir, the third (bronze hexagon lock nut) is located on the inlet line.
10. Clean the strainers, rinsing it with water.
11. Install the cleaned strainers.
12. Refill reservoir again with distilled water.
13. Switch on unit and observe/refill the reservoir level on a regular basis.
14. Install the top cover.
15. Continue with the start up procedure as described in the next paragraph.

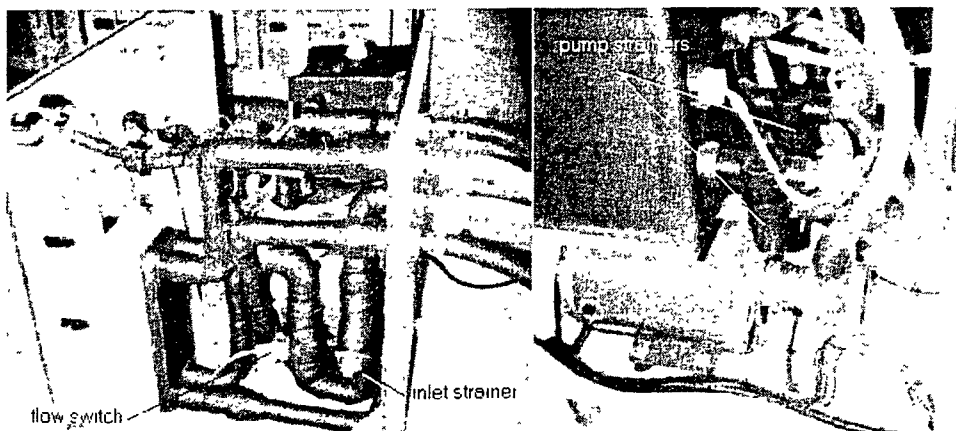


Figure 7 - Location of the three heat exchanger strainers

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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5. START UP PROCEDURE

NOTE

In case of faults during the following procedure consult the service manual supplied with the heat exchanger.

1. Verify the following:

- Temperature control knob is set to 25 °C.
- Facility cooling water must be connected. Note that the heat exchanger will regulate the amount of facility water depending on the temperature of the coil cooling water. The inlet facility water requirements are min. 20 l/min at 15 °C. Temperature range of inlet facility water is 8 to 15 °C.

2. Switch on the heat exchanger by pressing the POWER switch.

3. Check that:

- Recirculation pressure is between 43.5 PSI (3 bar) and 79.8 PSI (5.5 bar)
- Pressure fluctuation 7.2 PSI (0.5 bar)
- Recirculation temperature is about 25 °C.
- Verify/adjust the by-pass flow to 0.8 liter/minute.
The de-ioniser loop is equipped with a flow valve and flowmeter.
Open the access door on the front of the unit.
- After a short time the IDLE light must lit. This means that the temperature of the coil water is about 25 °C (set point).
- Purity light is green under normal condition.

CAUTION

*The heat exchanger must be switched ON continuously, also during the weekends.
This is necessary to keep up the ionisation level of (1 MΩ/cm) of the coil water circuit.
Switching off the heat exchanger for a long period will increase the ionisation level.
This will increase the electrolytic process and affect the gradient coil.*

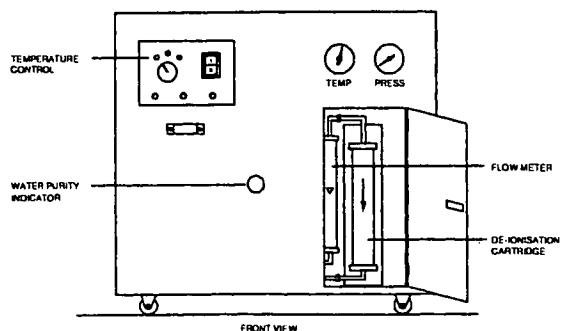


Figure 8 - Location of the flow meter

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6. APPENDIX: SETPOINT OF SECONDARY TEMPERATURE SWITCHES

On the secondary circuit there are two temperature switches which report the diagnostics of the heat exchanger. These switches are factory set. Check or adjust if required these settings as follows

Switch setting on coil side 'FEED line' : 90 ° F (30 °C)
 Switch setting on coil side 'RETURN line' : 100 ° F (35 °C)

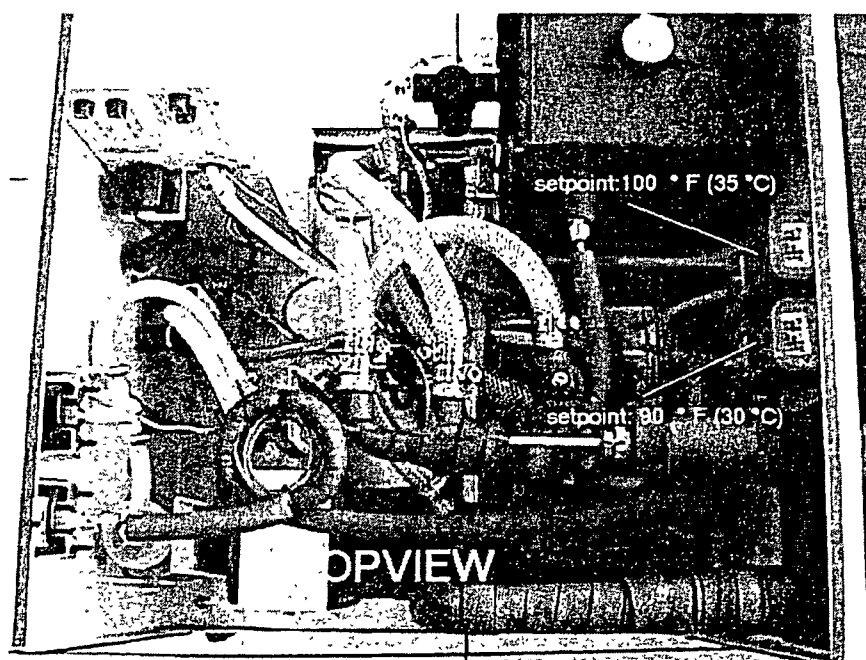


Figure 9 - Setpoint of temperature switches